Why Token Pruning Crashes Some Robot Policies but Not Others: Vision Connectors and VLA Token Compressibility

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Abstract: Vision token pruning accelerates Vision Transformer inference with negligible accuracy loss, but whether this robustness transfers to Vision-Language-Action (VLA) models remains untested across architectures. We study seven VLA models (450M–7B parameters) spanning three vision connector families, evaluating four in closed-loop robotic manipulation (LIBERO) and analyzing three via offline metrics or literature. The vision connector is strongly associated with pruning behavior along a four-class compressibility spectrum. Uniformly critical models (π_0, $\pi_0.5$; CV < 0.06) tolerate random subsampling but collapse under importance-based selection: $\pi_0.5$ retains 69.4% success rate with random removal at 50% retention but drops to 33.4% under L2 norm selection, a 36 percentage-point gap demonstrating that offline pruning tolerance does not guarantee closed-loop robustness. Threshold-graded models (X-VLA; CV $= 0.151$) maintain $97.8 \pm 0.4\%$ through 90% retention before cliffing at 85% (-31 pp). Smooth graded models (OpenVLA, OFT; CV ≈ 0.22) degrade proportionally. Catastrophic cliff models (SmolVLA; CV $= 0.314$) collapse to 0% at 5% removal across all seven tested configurations. The coefficient of variation (CV) of token L2 norms serves as a zero-cost diagnostic metric that correlates with both the tolerance class and the optimal pruning strategy from a single forward pass. The gap between random and L2 norm selection scales monotonically with CV at both 50% and 70% retention across three models. For catastrophic-cliff architectures, Vision Token Re-encoder (VTR) recovers 60%+ success rate at 75% compression where all deletion methods yield 0%, with a relatively stable M-curve (M=16 $\approx$ M=128) confirming learned re-encoding as the key factor. CV diagnosis followed by connector-appropriate compression forms a diagnostic protocol for safe VLA token reduction.

Keywords: vision-language-action models, token pruning, robot learning

1 Introduction

Vision token pruning reliably accelerates Vision Transformers with negligible accuracy loss [1, 2, 3]. Vision-Language-Action (VLA) models [4, 5, 6] use the same ViT encoders and produce hundreds of vision tokens that dominate the language backbone's input, making token reduction attractive for real-time robotic control at 5–10 Hz. Yet applying pruning to VLAs yields wildly inconsistent results: SmolVLA [6] collapses to 0% task success at 5% token removal, while π_0 [5] shows a 2.4% *improvement* in offline action loss at 90% removal (offline metrics only; no closed-loop evaluation available). No existing work explains which outcome a given architecture will produce.

The critical architectural variable is the vision connector that transforms ViT patches into tokens for the language backbone. Pixel-shuffle connectors [6, 7] fold spatial patches so each token en-

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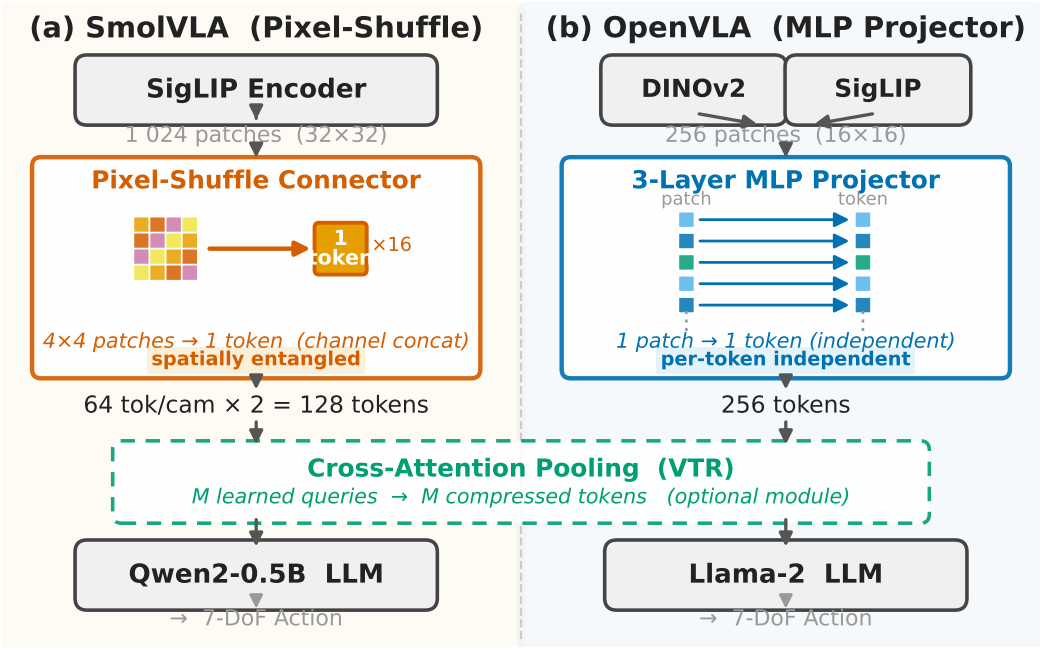


Figure 1: Vision connector comparison. SmolVLA's pixel-shuffle folds 4×4 patches into single tokens (128 total, spatially entangled). OpenVLA's MLP projector maps each patch independently (256 tokens, per-token independence). The connector design is strongly associated with token compressibility.

codes a 4×4 receptive field. MLP projectors [8, 4] map patches independently through learned nonlinearities. Linear projections [5, 9] apply a single matrix multiplication. Recent VLA-specific works [10, 11, 12] evaluate individual architectures without testing whether pruning behaves consistently across connector designs (Figure 1). Unlike static benchmarks, robotic manipulation compounds errors across timesteps, making compression failures far more consequential.

We study vision token pruning across seven VLA models (450M–7B) spanning three connector families on LIBERO [13], evaluating four in closed-loop and analyzing three via offline metrics or literature. The vision connector is strongly associated with pruning behavior along a four-class spectrum, and the coefficient of variation (CV) of token L2 norms correlates with both the tolerance class and the optimal strategy from a single forward pass. A central finding: low-CV models are not immune to pruning but *uniformly critical*. Closed-loop evaluation of $\pi_0.5$ [9] (CV = 0.055) reveals a 36 pp gap between random and L2 norm selection at k=0.5, demonstrating that models showing offline pruning tolerance can exhibit severe closed-loop degradation when the pruning method introduces spatial bias. CV provides a zero-cost diagnostic: a single forward pass replaces hundreds of expensive closed-loop episodes.

Our contributions:

1. **Connector-associated compressibility taxonomy with design guidelines** (§4): Four classes—uniformly critical (linear projection), threshold-graded (linear + deep post-processing), smooth graded (MLP), catastrophic cliff (pixel-shuffle)—with practical guidelines for each. The monotonic CV-degradation relationship across seven models is consistent with the connector being a primary factor, though confounded by backbone differences (Appendix N).
2. **CV as diagnostic metric** (§4.6, Appendix K): Token importance CV correlates with both tolerance class and optimal pruning method from a single forward pass. Δ(R−L2) at k=0.5 scales monotonically from +36.0 pp (CV = 0.055) to −1.8 pp (CV = 0.215), validated at both k=0.5 and k=0.7.

3. **VTR overcomes catastrophic cliff** (§4.5, Appendix L): Vision Token Re-encoder recovers 0% $\rightarrow$ 60%+ SR at 75% compression via learned re-encoding (cross-attention and MLP adapter, $p = 0.15$; relatively stable M-curve with $M{=}16 \approx M{=}128$).

2 Related Work

Vision token efficiency. Token pruning in ViTs removes 30–60% of tokens with $<$1% accuracy loss on classification, via learned masks [1], attention-guided reorganization [3], bipartite matching [2], window-level sparsity [14], adaptive halting [15, 16], or learned spatial selection [17]. For VLMs, pruning extends to the language backbone: FastV [10] removes tokens after early LLM layers, LLaVA-PruMerge [18] and FasterVLM [19] use CLS-token attention as an importance signal, FitPrune [20] minimizes attention-distribution divergence without training, and CrossGET [21] applies cross-modal token merging; TokenPacker [22] re-encodes compressed representations. All evaluate on classification or VQA benchmarks; we test whether these results transfer to closed-loop VLA control, where action errors compound across timesteps.

VLA-specific token reduction. VLA-Cache [11] caches vision KV states across timesteps for 1.7$\times$ speedup without removing tokens. VLA-IAP [12] uses interaction-aligned pruning on OpenVLA; EfficientVLA [23] combines layer pruning and token selection on CogACT [24]; VLA-Pruner [25] uses dual-level importance criteria for temporal-aware pruning. All evaluate on MLP-projector architectures. None reports the cliff-edge failure we observe on pixel-shuffle VLAs; their positive results may not generalize to spatially entangled connectors.

Vision connectors. Linear projections [5, 9] apply no cross-token mixing, producing independently addressable tokens; $\pi_{0.5}$ [9] adds a learned $\times\sqrt{d}$ scaling that preserves near-uniform norms. Per-patch MLP projectors [8, 26, 4] preserve per-token granularity but introduce partial importance differentiation through nonlinear transformations. Cross-attention connectors (Q-Former [27], Perceiver [28, 29], RoboFlamingo [30]) compress to learned queries at the architectural level. Pixel-shuffle connectors [31, 6, 7] fold $k{\times}k$ spatial patches into channel-concatenated tokens, achieving training-time compression but creating inter-token spatial dependencies; InternVL 2.5 [32] and Mini-InternVL [33] scale this design from 1B to 78B parameters, and compact VLAs such as TinyVLA [34] adopt similar connectors. Our analysis links this taxonomy to inference-time compressibility: linear projections create uniform criticality where method selection matters more than retention ratio, MLP projectors permit graded pruning, and pixel-shuffle requires learned re-encoding.

VLA compression. Concurrent works compress tokens on MLP-projector VLAs: Compressor-VLA [35] uses instruction-guided compression with semantic and spatial modules, SpecPrune-VLA [36] combines self-speculative decoding with action-aware pruning, LightVLA [37] learns differentiable pruning masks during training, and "See What Matters" [38] applies differentiable grid sampling for generalizable pruning. SparseVLM [39] and SparseVILA [40] explore text-guided visual sparsification for VLM inference. "Don't Run with Scissors" [41] provides a complementary perspective: rather than token reduction, they study *weight* pruning and quantization on VLA models, and find that standard magnitude- and gradient-based pruning causes drastic performance degradation and increased safety violations. Their proposed recovery method, GLUESTICK, interpolates between dense and pruned weight spaces without additional training. This finding parallels our observation that naive compression breaks VLA control, though the failure mechanisms differ: weight pruning degrades learned representations uniformly across the network, while vision token deletion selectively removes spatial information with consequences that depend on how the connector entangles tokens. All of the above works operate on MLP-projector architectures or evaluate without diagnosing connector-dependent failure modes. We identify why deletion catastrophically fails on pixel-shuffle architectures and validate re-encoding as a remedy.

3 Methods and Experimental Setup

3.1 Architectures and Benchmark

Seven VLA architectures spanning three vision connector families provide the testbed (Table 1).

Linear projection. π_0 [5] (3.3B, 256 tokens) maps SigLIP [42] patches through a single linear layer; $\pi_0.5$ [9] (4B, 768 tokens from three cameras) adds a learned $\times\sqrt{d}$ scaling; X-VLA [43] (880M, 50 tokens) augments its linear projection with a 12-layer BART encoder [44] that introduces cross-token mixing. **MLP projector.** OpenVLA [4] (7B, 256 tokens) fuses DINOv2 [45] and SigLIP through a three-layer MLP; OFT [46] shares this connector (entropy comparison only); CogACT [24] retains 74.2% SR under 78% pruning per EfficientVLA [23] (literature corroboration, not our evaluation). **Spatial folding.** SmolVLA [6] (450M, 128 tokens) applies pixel-shuffle (4×4 folding) so each token encodes 16 contiguous patches.

We evaluate on LIBERO [13] (Object: 10 tasks $\times$ 50 episodes = 500 total; Spatial and Goal for cross-suite validation). Closed-loop success rate (SR) with Wilson 95% CIs is the primary metric; multi-seed results report mean $\pm$ std. Baselines: SmolVLA 44.8%, OpenVLA 74.0%, X-VLA 99.2%, $\pi_0.5$ 98.2%.

3.2 Pruning Protocol

Two methods bracket robustness. **L2 norm-based removal** ranks tokens by L2 norm and physically removes the lowest-norm fraction, providing a favorable test. **Random removal** selects tokens uniformly at random, providing a conservative lower bound. Both physically remove tokens (not zero-masking); remaining tokens retain spatial ordering with re-indexed position IDs. Per-model retention ratios and cross-architecture comparison details appear in Appendix A.

3.3 Vision Token Re-encoder (VTR)

A cross-attention module (22.2M parameters, $\sim$5% of SmolVLA) is inserted between the pixel-shuffle connector and the LLM backbone. M learned queries attend over 128 source tokens, producing M compressed representations ($M \in \{16, 32, 64, 96, 128\}$; M=128 = passthrough control). A parameter-matched MLP adapter (22.14M) serves as architecture control. Both train from SmolVLA-Base for 100K steps with identical hyperparameters; an extended training control without the compressor achieves 45.0% SR ($p > 0.9$ vs. 44.8% baseline). Training details in Appendix F.

3.4 Diagnostic Metrics

Token importance CV (coefficient of variation of L2 norms) tracks pruning tolerance continuously: uniformly critical at CV < 0.06, threshold-graded at $\sim$0.15, smooth graded at $\sim$0.22, catastrophic cliff at > 0.3. **Attention entropy ratio** (normalized Shannon entropy of attention weights over vision tokens) provides a complementary view: OFT and OpenVLA share nearly identical entropy (0.667 vs. 0.661), confirming the connector is associated with token utilization patterns independent of action head.

4 Results

We evaluate token pruning across six VLA models spanning three connector families on LIBERO-Object [13] (10 tasks, 50 episodes each, 500 total). All success rates report Wilson 95% confidence intervals for single-seed results and mean $\pm$ standard deviation for multi-seed results.

4.1 Taxonomy Overview

Table 1 organizes seven models into four response classes. The taxonomy reveals that effective vision-to-LLM pipeline complexity, not the nominal connector label or model scale, is strongly

Table 1: Seven-model connector taxonomy. CV = coefficient of variation of vision token L2 norms. $^{\dagger}\pi_0$: offline metrics only, no closed-loop evaluation; the 63 pp gap vs. OpenVLA at k=0.1 mixes offline loss with closed-loop SR. ‡X-VLA uses a linear projection followed by a 12-layer BART encoder; the effective pipeline is not purely linear. §OFT reports attention entropy and CV only (no pruning evaluation); both metrics match OpenVLA, consistent with the connector being associated with token importance structure. $^{\|}$CogACT: literature result from EfficientVLA [23], not our evaluation.

Model	Connector	Params	Tokens	CV	Representative Result	Class
π_0 [5]	Linear proj	3.3B	256	0.038	k=0.1: $\mathcal{L}_1$=0.976 (+2.4%)†	Uniformly critical
$\pi_0.5$ [9]	Linear proj ($\times\sqrt{d}$)	4B	768	0.055	k=0.5: L2=33.4%, R=69.4% (Δ=+36pp)	Uniformly critical
X-VLA [43]	Linear + BART (12L)‡	880M	50	0.151	k=0.9: 97.8±0.4, k=0.85: 67.9±0.3	Threshold-graded
OpenVLA [4]	3-layer MLP	7B	256	0.215	k=0.5: 37.4% SR (−49%)	Smooth graded
OFT [46]	3-layer MLP	7B	256	0.219	CV 0.219 ≈ 0.215; Ent. 0.667 ≈ 0.661§	Smooth graded
CogACT [24]	MLP proj	–	–	–	74.2% SR under 78% prune$^{\|}$	Smooth graded
SmolVLA [6]	Pixel-shuffle	450M	128	0.314	k=0.95: 0% SR	Catastrophic cliff

associated with pruning tolerance. Token importance CV tracks this ordering continuously across an 8× parameter range (450M to 7B): uniformly critical models cluster at CV < 0.06, threshold-graded at ∼0.15, smooth graded at ∼0.22, and catastrophic cliff at >0.3. A pairwise comparison between π_0 and OpenVLA (both consuming 256 SigLIP-based [42] tokens but differing in connector, backbone, and encoder fusion) yields a 63 pp gap at k=0.1; this gap is consistent with the monotonic CV-degradation pattern across all seven models (Section 4.4).

The four classes differ qualitatively in their failure mode. Uniformly critical models (π_0, $\pi_0.5$ [9]) distribute importance so evenly that no scoring method can identify dispensable tokens; pruning either preserves random spatial coverage (and survives) or removes a clustered patch (and fails). Threshold-graded models (X-VLA) tolerate removal up to a sharp onset point. Smooth graded models (OpenVLA, OFT, CogACT) degrade proportionally to the fraction removed. Catastrophic cliff models (SmolVLA) collapse at any removal level.

4.2 Pixel-Shuffle: Catastrophic Cliff-Edge

The SmolVLA baseline (bypass mode, all 128 vision tokens) achieves 44.8% SR (224/500, CI [40.5%, 49.2%]).[1] Every pruning configuration produces 0% SR across seven configurations varying scoring (learned ACIS vs. random), retention ratio ($k \in \{0.50, 0.70, 0.90, 0.95\}$), and mechanism (zero-masking, true removal, token merging). The cliff persists regardless of checkpoint quality: the official SmolVLA HF checkpoint (57.2% SR unpruned, CI [52.9%, 61.4%]) also collapses to 0% under random k=0.95 (5% removal), L2 norm k=0.9 (10% removal), and L2 norm k=0.5 (Table 3). The failure is architectural, not a training artifact.

The pattern generalizes across LIBERO suites: Spatial 0.6%, Goal 0%. Pixel-shuffle folds each 4×4 patch block into a single token by channel concatenation; removing one token eliminates an entire 4×4 receptive field. The resulting spatial entanglement means even 5% removal (k=0.95, losing 6 of 128 tokens) destroys task-critical visual information. Full ablation results appear in Table 3, Appendix B.

4.3 MLP Projector: Smooth Graded Degradation

Under random token removal, OpenVLA degrades gracefully across all retention ratios: 74.0% (baseline) → 69.4% at k=0.95 (−4.6 pp) → 62.0% at k=0.90 → 49.0% at k=0.70 → 37.4% at k=0.50 → 20.6% at k=0.30 (Figure 2a; full results with CIs in Table 7, Appendix).[2] At k=0.30,

[1] The official SmolVLA result is 87.3% SR with batch size 64 on 8 GPUs. Our reproduction uses a smaller batch due to compute constraints.

[2] A second random seed confirms the graded pattern: 62.2% at k=0.9 (+0.2 pp), 44.4% at k=0.7 (−4.6 pp), 35.6% at k=0.5 (−1.8 pp). All within 5 pp of the primary seed. Per-seed details in Table 10, Appendix C.9.

two tasks reach 0% SR, but eight retain non-zero performance: degradation is graded, not cliff-edge. Random token removal provides a conservative lower bound on architectural resilience; learned scoring methods would likely yield better retention.

OFT [46] retains OpenVLA's MLP connector but replaces the action head; its attention entropy (0.667) matches standard OpenVLA (0.661). The connector, not the action head, is associated with token importance structure. EfficientVLA [23] corroborates the graded pattern from the literature: CogACT [24] (MLP projector) retains 74.2% SR under 78% pruning.

To control for scoring method in cross-architecture comparisons, we evaluate L2 norm-based selection on OpenVLA. At moderate-to-aggressive retention ($k \leq 0.7$), L2 performs within 2–6 pp of random removal[3] (Table 11, Appendix C.10). At k=0.9, L2 outperforms random by 8.6 pp (70.6% vs. 62.0%, reproducible across two random seeds), consistent with the weak but non-zero importance gradient predicted by CV = 0.215: the signal is exploitable only when few tokens are removed.

4.4 Linear Projection: From Uniformly Critical to Threshold-Graded

Three models share a linear projection connector yet exhibit qualitatively different pruning responses. The divergence traces to what happens *after* the projection: nothing (π_0), a learned scaling ($\pi_0.5$), or a full 12-layer encoder (X-VLA).

π_0 (3.3B, 256 tokens, pure linear). Offline normalized L1 action loss at k=0.1 is 0.976, a 2.4% *improvement* over the unpruned baseline.[4] CV = 0.038 indicates near-uniform token norms. Because all tokens carry equal weight, removing 90% at random preserves spatial coverage and the model compensates.

$\pi_0.5$ (4B, 768 tokens, linear $\times\sqrt{d}$). The baseline achieves 98.2% SR. This model reveals that low CV does not imply tokens are dispensable; it implies uniform criticality. The response is strongly method-dependent: random removal at k=0.5 retains 69.4% SR (-28.8 pp), while L2 norm-based removal at the same ratio collapses to 33.4% (-64.8 pp). The gap of 36 pp is the largest method-dependent effect at k=0.5 in our study.

At $k \leq 0.7$, random removal degrades monotonically (77.2%, 69.4%, 36.4% at k=0.7, 0.5, 0.3; seed43 within 1 pp). At higher retention, a task-specific random cliff emerges on LIBERO-Object: SR drops from 69.4% (k=0.95) to 5.0% (k=0.85) before recovering to 77.2% (k=0.7), a 72 pp swing; the same k=0.85 on LIBERO-Spatial retains 58.4% (Appendix C.6). L2 norm selection produces a V-shaped curve: 55.0% at k=0.7, 33.4% at k=0.5 (the valley), 51.0% at k=0.3, confirmed across two seeds ($\sigma = 0.1$ pp at the valley).[5] The non-monotonic L2 pattern arises because norm ranking in a low-CV distribution selects spatially clustered tokens (slight norm variations correlate with image regions), and the resulting loss of spatial coverage peaks at intermediate budgets.

This result demonstrates that models showing offline pruning tolerance can exhibit severe closed-loop degradation. Low-CV tokens are uniformly critical, not uniformly dispensable. L2 norm-based selection systematically concentrates on spatially correlated tokens and loses coverage; random selection preserves the uniform spatial distribution that these architectures require.

X-VLA (880M, 50 tokens, linear + 12-layer BART). The 99.2% baseline is maintained at k=0.95 (99.1 $\pm$ 0.2%, 3 seeds) and k=0.9 (97.8 $\pm$ 0.4%) under L2 norm-based removal. A sharp cliff appears at k=0.85 (67.9 $\pm$ 0.3%, -31 pp, 3 seeds), with non-monotonic recovery at k=0.80 (89.0 $\pm$ 0.7%, 3 seeds) before degradation resumes: 58.6 $\pm$ 0.4% at k=0.7, 43.4 $\pm$ 0.8% at k=0.5,

[3] OpenVLA random results use seed = 7; L2 norm results use seed = 42. Same-seed (seed42) gaps: 5.4 pp at k=0.7, 3.6 pp at k=0.3 (21.4% random vs. 25.0% L2); same-seed at k=0.5: 35.8% random vs. 39.2% L2 (gap 3.4 pp).

[4] Offline action loss measures single-step prediction quality; closed-loop evaluation may reveal additional degradation.

[5] V-shape valley at k=0.5: 33.4% (seed42) vs. 33.6% (seed43), σ=0.1 pp. V-depth: 17.6 pp (seed42), 21.8 pp (seed43). Full multi-seed table in Appendix C.5.

Table 2: VTR re-encoding vs. deletion on SmolVLA (LIBERO-Object). The M-curve is relatively stable: M=16 (87.5% compression) matches M=128 (no compression) within 1.1 pp. A parameter-matched MLP adapter achieves comparable SR ($p = 0.15$ vs. cross-attention). ‡Mean ± std over 3 seeds (500 episodes each). Other rows: single seed with Wilson 95% CI.

Method	M	Red.	SR (%)	Seeds
Baseline	128	0%	44.8	1
Ext. training	–	–	45.0	1
Hard pruning	–	50–90%	0.0	mult.
Cross-attn‡	16	87.5%	56.9±1.3	55.8/58.4/56.5
Cross-attn‡	32	75%	60.1±1.3	61.6/59.2/59.6
Cross-attn‡	64	50%	**62.0**±2.3	61.6/60.0/64.4
Cross-attn‡	96	25%	54.5±3.2	52.6/58.2/52.8
Cross-attn‡	128	0%	58.0±3.5	59.8/54.0/60.2
MLP adapter‡	32	75%	63.4±2.6	65.2/64.6/60.4
Official checkpoint (59.2% baseline)				
Cross-attn	32	75%	53.4	1

$22.0 \pm 0.9\%$ at k=0.3, $4.7 \pm 0.6\%$ at k=0.2, and near-zero at k=0.1. The cliff at k=0.85 (retaining 42 of 50 BART-encoded tokens) is reproducible across seeds (std $\leq$ 1.6 pp). The non-monotonic dip is task-specific: four tasks lose ∼50 pp each while six remain unaffected (Table 12, Appendix C.11). Under random removal, X-VLA exhibits a plateau: $65.1 \pm 0.1\%$ at k=0.7 and $64.0 \pm 1.6\%$ at k=0.3 (3 seeds each), confirming that L2 norm-based selection ($22.0 \pm 0.9\%$ at k=0.3) is far more destructive than random for this intermediate-CV architecture.

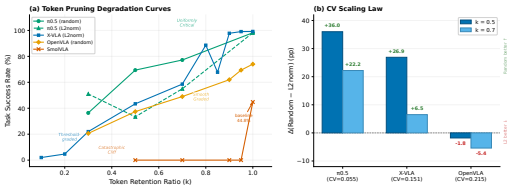
That this threshold appears under importance-based removal (the more favorable test) while OpenVLA degrades smoothly under random removal (the harder test) strengthens the architecture-dependent conclusion. The π_0–OpenVLA comparison (both consuming 256 SigLIP-based tokens, differing in connector, backbone, and encoder fusion) produces a 63 pp gap at k=0.1, consistent with the connector-associated pattern observed across all seven models. This comparison involves differences in connector, backbone, and encoder fusion, limiting causal interpretation.

4.5 VTR: Re-Encoding Overcomes Cliff-Edge

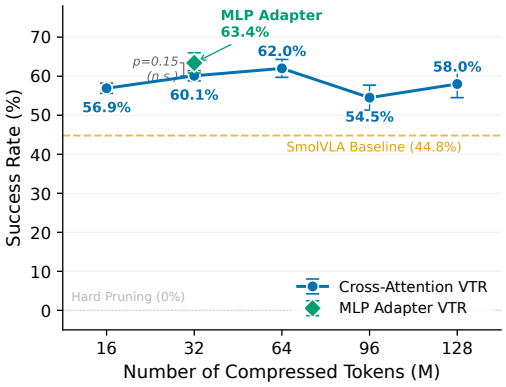
To test whether learned re-encoding can overcome the pixel-shuffle cliff-edge, we insert a 22.2M-parameter cross-attention module (Vision Token Re-encoder, VTR) between the vision connector and the LLM backbone (details in Appendix F). VTR compresses 128 input tokens to M output tokens. Both VTR and the baseline train from the same SmolVLA-Base checkpoint for 100K steps with identical hyperparameters. An extended training control (same procedure without the compressor) achieves 45.0% SR ($p > 0.9$ vs. the 44.8% baseline); the gain is architectural, not from additional optimization steps.[6]

The M-curve is relatively stable (Table 2): M=16 (56.9%, 87.5% compression) performs within 1.1 pp of M=128 (58.0%, no compression), and all five M values cluster within 8 pp. A parameter-matched MLP adapter (22.14M parameters) achieves 63.4% ± 2.6 at M=32 ($p = 0.15$ vs. cross-attention); both architectures achieve comparable recovery, suggesting learned re-encoding in general is the key (Figure 2b). Cross-suite evaluation confirms generalization: 53.2% on LIBERO-Spatial and 56.4% on LIBERO-Goal where hard pruning yields 0% (Appendix D). VTR applied to OpenVLA yields 0% SR even in passthrough mode (M=128): it is a targeted remedy for pixel-shuffle entanglement, not a universal compression module.

[6]The extended baseline also collapses to 0% SR under pruning (k=0.50, random, true removal). Additional training does not mitigate the pixel-shuffle vulnerability.



(a) Token pruning across connector classes.



(b) VTR M-curve and CV-strategy interaction.

Figure 2: Token reduction on LIBERO-Object. (a) Pruning: linear (π_0) uniformly critical, linear+encoder (X-VLA) threshold-graded, MLP (OpenVLA) smooth, pixel-shuffle (SmolVLA) collapses. (b) VTR: M=16 $\approx$ M=128 where deletion yields 0%; CV-gap monotonic.

4.6 CV Correlates with Optimal Pruning Strategy

Token importance CV correlates with both pruning tolerance and optimal strategy. Comparing L2 norm-based and random removal across three models, the gap Δ(R−L2) decreases monotonically with CV at both k=0.5 and k=0.7 (full data in Table 19, Appendix P). At k=0.5: +36.0 pp (π_0.5, CV = 0.055) $\rightarrow$ +26.9 pp (X-VLA, CV = 0.151) $\rightarrow$ −1.8 pp (OpenVLA, CV = 0.215). At k=0.7: +22.2 pp $\rightarrow$ +6.5 pp $\rightarrow$ −5.4 pp. The monotonic ordering holds across a 2$\times$ change in token budget, indicating CV captures a structural property of the connector. Below CV $\approx$ 0.15, L2 ranking captures spatial artifacts rather than importance; above it, importance-based scoring becomes beneficial (reversal at k=0.3; Appendix Q).

4.7 Cross-Suite Generalization

The cliff zone exhibits a suite $\times$ method interaction: at $k \in [0.85, 0.95]$, Object tasks collapse under random removal while Spatial tasks degrade gracefully (Figure 3, Table 9). L2 norm maintains a near-baseline plateau on both suites ($\leq$ 2.4 pp at $k \geq 0.85$). CV = 0.055 correctly predicts this plateau on both suites despite qualitatively different cliff profiles, confirming CV as a model-level diagnostic. Token importance heatmaps reveal a center-high, edge-low spatial pattern (Figure 4, Appendix C.7), formalized by a hypergeometric model $H(N$=768, C=91, T=6): cliff onset at k=0.96 (10% collapse probability) narrows to k=0.90 (90%), RMSE=0.025 (Appendix C.8)—driven by Object's concentrated criticality (C=91) versus Spatial's distributed pattern ($C \approx 5$).

5 Discussion

Re-encoding and CV diagnostics. Both cross-attention and MLP modules achieve comparable recovery ($p = 0.15$); the stable M-curve (M=16 $\approx$ M=128) confirms token count is secondary once entanglement is broken (Appendix L). CV captures effective pipeline complexity beyond connector labels: X-VLA shares a linear projection with π_0 yet its BART encoder yields CV = 0.151 and threshold-graded behavior, correctly differentiating it from π_0's CV = 0.038. The CV-strategy interaction partially reverses at k=0.3 (Appendices Q, K).

Limitations. All closed-loop experiments use LIBERO simulation; the seven models differ in backbone and encoder fusion, preventing clean causal isolation (Appendix N). π_0 is offline only; the CV correlation spans three models and two retention ratios; the hypergeometric cliff model captures onset sharpness (RMSE = 0.025) but does not explain V-shape recovery at $k \leq 0.7$.

6 Conclusion

We studied vision token pruning across seven VLA models (450M–7B) spanning three connector families on closed-loop LIBERO. The vision connector is strongly associated with pruning behavior along a four-class spectrum; token importance CV correlates with both tolerance class and optimal strategy. Low-CV models are uniformly critical, not immune ($\pi_0.5$: 36 pp gap at k=0.5); for pixel-shuffle architectures, learned re-encoding recovers 60%+ SR at 75% compression where deletion yields 0%.

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A Pruning Protocol Details

Both pruning methods physically remove unselected tokens from the sequence, reducing the actual input length to the LLM rather than zeroing them out; zero-masking preserves positional structure while physical removal does not. After removal, remaining tokens retain their original spatial ordering; position IDs are re-indexed contiguously.

Model-specific retention ratios reflect each architecture's token count and expected sensitivity. π_0 is evaluated offline only (k=0.1, normalized L1 action loss; offline metrics only, no closed-loop evaluation available) because its deployment environment precludes closed-loop testing. $\pi_0.5$ uses $k \in \{0.3, 0.5, 0.7\}$ under both L2 and random removal; the dual-method design enables direct comparison within a single architecture. Pruning is applied per-image: each camera view independently retains the top-k fraction. X-VLA spans $k \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95\}$ with L2 removal across three seeds, supplemented by random removal at selected k values. OpenVLA uses $k \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$ with random removal as the primary protocol and L2 for direct comparison. SmolVLA tests seven configurations varying scoring (learned ACIS vs. random), retention ratio ($k \in \{0.5, 0.7, 0.9, 0.95\}$), and mechanism (zero-masking, true removal, token merging via ToMe [2]).

The π_0–OpenVLA pair provides the closest available comparison: both consume 256 vision tokens from SigLIP-based encoders, but differ in connector (linear projection vs. three-layer MLP), backbone (PaliGemma vs. Llama-2), and encoder fusion strategy (single SigLIP vs. DINOv2+SigLIP). This comparison involves differences in connector, backbone, and encoder fusion, limiting causal interpretation; the primary evidence for the connector's role comes from the monotonic CV-degradation relationship across all seven models (Section 4.6). Retention ratio k controls the fraction retained within each model, but absolute token counts vary across architectures at matched k: k=0.7 retains 35 tokens for X-VLA, 179 for OpenVLA, and 538 for $\pi_0.5$. The CV analysis derives from within-model comparisons where each model serves as its own control, eliminating this confounder.

B SmolVLA Pruning Ablation

C Per-Task Breakdown Tables

C.1 SmolVLA Baseline Per-Task Results

Table 4 reports per-task success rates for the SmolVLA baseline (bypass mode, no pruning) on LIBERO-Object. All pruning conditions (Table 3) produce 0% success rate on every task; we omit per-task breakdowns for pruning conditions as they carry no additional information beyond the aggregate 0% already reported.

C.2 Official SmolVLA Checkpoint Per-Task Results

Table 5 reports per-task success rates for the official SmolVLA LIBERO checkpoint (HuggingFaceVLA/smolvla_libero). Under 50% random token removal, all tasks produce 0% SR, identical to the fine-tuned checkpoint results.

Table 3: SmolVLA pruning ablation on LIBERO-Object. All configurations collapse to 0% SR. Two independent evaluations of the official HF checkpoint (59.2% and 57.2% baseline, within CI) confirm the cliff-edge across scoring methods.

Scoring	k	Mechanism	SR (%)	Episodes	Wilson 95% CI
Baseline (bypass)		–	44.8	500	[40.5, 49.2]
ACIS	0.50	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.70	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.90	Zero-mask	0.0	500	[0, 0.76]
Random	0.95	Zero-mask	0.0	500	[0, 0.76]
Random	0.50	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.50	True removal	0.0	500	[0, 0.80]
ACIS	0.50	Merge (ToMe)	0.0	500	[0, 0.76]
Official checkpoint—eval protocol A (59.2% baseline)					
Baseline (bypass)		–	59.2	500	[54.8, 63.4]
Random	0.50	True removal	0.0	500	[0, 0.76]
Official checkpoint—eval protocol B (57.2% baseline)					
Baseline (bypass)		–	57.2	500	[52.9, 61.4]
Random	0.95	True removal	0.0	500	[0, 0.76]
L2 norm	0.90	True removal	0.0	500	[0, 0.76]
L2 norm	0.50	True removal	0.0	500	[0, 0.76]

Table 4: SmolVLA baseline (bypass mode) per-task success rates on LIBERO-Object (50 episodes per task, 500 total). Overall SR: 44.8%.

Task	SR (%)
alphabet_soup	22
bbq_sauce	60
butter	42
chocolate_pudding	38
cream_cheese	58
ketchup	26
milk	62
orange_juice	38
salad_dressing	50
tomato_sauce	52
Overall	**44.8**

C.3 OpenVLA Per-Task Results Under Pruning

C.4 OpenVLA Full Degradation Curve

C.5 $\pi_0.5$ L2 Norm Multi-Seed Consistency

C.6 $\pi_0.5$ Cross-Suite Comparison

The random cliff at $k \in [0.85, 0.925]$ is task-specific: Object collapses from 69.4% (k=0.95) to 5.0% (k=0.85), whereas Spatial retains 58.4% at k=0.85 (Δ=53 pp between suites). Spatial random also exhibits mild non-monotonicity (k=0.925: 60.4% $>$ k=0.9: 54.6%, Δ=5.8 pp), but this is an order of magnitude smaller than Object's 65 pp V-dip. The cliff pattern suggests Object manipulation concentrates critical information in specific token subsets whose removal is catastrophic at narrow retention bands; Spatial tasks distribute critical information more uniformly, tolerating moderate random removal without catastrophic failure.

L2 norm selection is consistent across suites at high retention ($k \geq 0.85$): Object and Spatial differ by ≤ 2.4 pp (Table 9). At $k \leq 0.7$, both suites exhibit the L2 V-shape (Object valley: 33.4% at k=0.5; Spatial valley: 55.6% at k=0.5), with Spatial maintaining higher absolute SR throughout.

Table 5: Official SmolVLA checkpoint per-task success rates on LIBERO-Object (eval protocol A; 50 episodes per task, 500 total). Overall SR: 59.2%. Under 50% random pruning, all tasks collapse to 0%. An independent evaluation (protocol B) yields 57.2% baseline with identical cliff-edge (Table 3).

Task	Baseline SR (%)	k=0.5 SR (%)
alphabet_soup	64	0
bbq_sauce	38	0
butter	78	0
chocolate_pudding	90	0
cream_cheese	40	0
ketchup	60	0
milk	52	0
orange_juice	80	0
salad_dressing	44	0
tomato_sauce	46	0
Overall	**59.2**	**0.0**

Table 6: OpenVLA per-task success rates (%) on LIBERO-Object under random token removal across five retention ratios (50 episodes per task). Baseline overall SR: 74.0%. Tasks requiring fine-grained discrimination (butter, cream_cheese) degrade earliest; visually distinctive targets (orange_juice, salad_dressing) are most resilient.

Task	k=0.95	k=0.90	k=0.70	k=0.50	k=0.30
alphabet_soup	76	70	74	60	14
bbq_sauce	38	28	46	44	30
butter	84	76	10	2	0
chocolate_pudding	54	56	32	40	10
cream_cheese	84	78	18	8	0
ketchup	82	70	68	50	30
milk	46	42	40	24	8
orange_juice	74	60	76	72	42
salad_dressing	64	54	64	36	64
tomato_sauce	92	86	62	38	8
Overall	**69.4**	**62.0**	**49.0**	**37.4**	**20.6**

The method preference (random $>$ L2 at k=0.5; L2 $>$ random at k=0.3) is consistent across the two suites tested, suggesting the CV-strategy interaction may extend beyond a single task distribution.

C.7 Token Importance Spatial Distribution

Token importance heatmaps reveal that $\pi_0.5$'s 768 vision tokens (reshaped to a 16×16 spatial grid after the $\sqrt{d}$-scaled linear projection) exhibit a center-high, edge-low L2 norm pattern. At k=0.9 (26 tokens removed), L2 norm-based selection discards predominantly edge tokens: 50% of removed tokens fall in the outermost ring despite edge positions comprising only 23% of the grid. This spatial concentration is consistent with the center tokens encoding the robot arm and target object, while edge tokens capture less task-relevant background.

Random removal at the same ratio samples deletions uniformly across the spatial grid, creating a combinatorial probability of simultaneously removing multiple high-importance center tokens. As more tokens are removed (k=0.925 $\rightarrow$ 0.85), this probability increases sharply, consistent with the observed cliff onset. The asymmetry explains why L2 norm selection maintains a near-baseline plateau through the cliff zone: it preferentially preserves the high-importance center region that random removal disrupts by chance.

Table 7: OpenVLA success rates on LIBERO-Object under random token removal at six retention ratios (50 episodes per task, 500 total per ratio). Baseline SR: 74.0%. All confidence intervals are Wilson 95%.

k	Tokens Retained	SR (%)	Wilson 95% CI
1.00 (baseline)	256	74.0	[69.9, 77.7]
0.95	243	69.4	[65.2, 73.3]
0.90	230	62.0	[57.7, 66.2]
0.70	179	49.0	[44.6, 53.4]
0.50	128	37.4	[33.3, 41.7]
0.30	77	20.6	[17.2, 24.4]

Table 8: $\pi_0.5$ L2 norm-based removal across two seeds (500 episodes each). The V-shaped curve is reproduced with near-identical valleys (σ=0.1 pp at k=0.5). V-depth: 17.6 pp (seed42), 21.8 pp (seed43).

k	Seed 42 (%)	Seed 43 (%)	Mean $\pm$ Std	Δ (pp)
0.70	55.0	57.8	56.4 ± 2.0	+2.8
0.50	33.4	33.6	33.5 ± 0.1	+0.2
0.30	51.0	55.4	53.2 ± 3.1	+4.4

C.8 Hypergeometric Cliff Model

The random cliff on $\pi_0.5$ Object (Section 4.7) can be modeled as a hypergeometric sampling problem. Among N=768 vision tokens, C=91 are critical—defined as tokens whose L2 norm exceeds the 87.5th percentile, corresponding to the center-high region identified in the importance heatmap (Appendix C.7). When $\lfloor N(1-k) \rfloor$ tokens are removed uniformly at random, the number of critical tokens removed follows a hypergeometric distribution $\mathrm{Hypergeometric}(N, C, \lfloor N(1-k) \rfloor)$. We define performance collapse as removing $\geq T$=6 critical tokens simultaneously; fitting T to minimize RMSE against observed success rates in the cliff zone yields T=6 with RMSE=0.025.

The model predicts three characteristic retention ratios: k=0.963 where $P(\text{collapse})$=10% (cliff onset), k=0.938 where $P(\text{collapse})$=50% (midpoint), and k=0.903 where $P(\text{collapse})$=90% (near-complete collapse). The cliff width $\Delta k \approx 0.06$ between 10% and 90% collapse probability matches the observed transition from 69.4% SR (k=0.95) to 12.0% SR (k=0.90). The model captures the cliff onset and sharpness but does not explain the V-shape recovery at $k \leq 0.7$, where success rate rebounds to 77.2%; characterizing the recovery mechanism is left to future work.

The cross-suite difference emerges naturally from the critical token count. Object manipulation concentrates critical information in C=91 tokens (12.5% of the grid), creating a steep increase in $P(\text{collapse} \geq T)$ over a narrow k range. Spatial tasks distribute critical information across approximately $C \approx 5$ tokens, making the probability of simultaneously removing $\geq T$ critical tokens negligible at all tested retention ratios—consistent with Spatial's graceful degradation (58.4% at k=0.85 vs. Object's 5.0%).

C.9 OpenVLA Multi-Seed Consistency

C.10 L2 Norm vs. Random Scoring on OpenVLA

C.11 X-VLA Cliff Localization: Per-Task Analysis

The non-monotonic pattern is reproducible across both seeds: the same four tasks (butter, chocolate_pudding, cream_cheese, ketchup) exhibit catastrophic drops at k=0.85, while the remaining six maintain or improve performance. This task-specific sensitivity is consistent with the BART encoder creating entangled dependencies among marginal tokens: at k=0.85, two additional low-importance tokens (L2 rank 41–42) introduce interference that selectively disrupts tasks requiring fine-grained

Table 9: $\pi_0.5$ pruning across LIBERO-Object (baseline 98.2%) and LIBERO-Spatial (baseline 97.2%), seed 42, 500 episodes each. Object random exhibits a deep cliff at $k \in [0.85, 0.925]$ absent on Spatial; L2 norm maintains a plateau across suites at high retention (≤ 2.4 pp difference at $k \geq 0.85$).

k	LIBERO-Object (%)		LIBERO-Spatial (%)	
	L2 Norm	Random	L2 Norm	Random
1.00	98.2	98.2	97.2	97.2
0.95	98.2	69.4	97.8	76.4
0.925	99.2	27.6	97.0	60.4
0.90	98.2	12.0	95.0	54.6
0.85	98.0	5.0	90.4	58.4
0.70	55.0	77.2	67.0	74.4
0.50	33.4	69.4	55.6	71.6
0.30	51.0	36.4	57.2	49.6

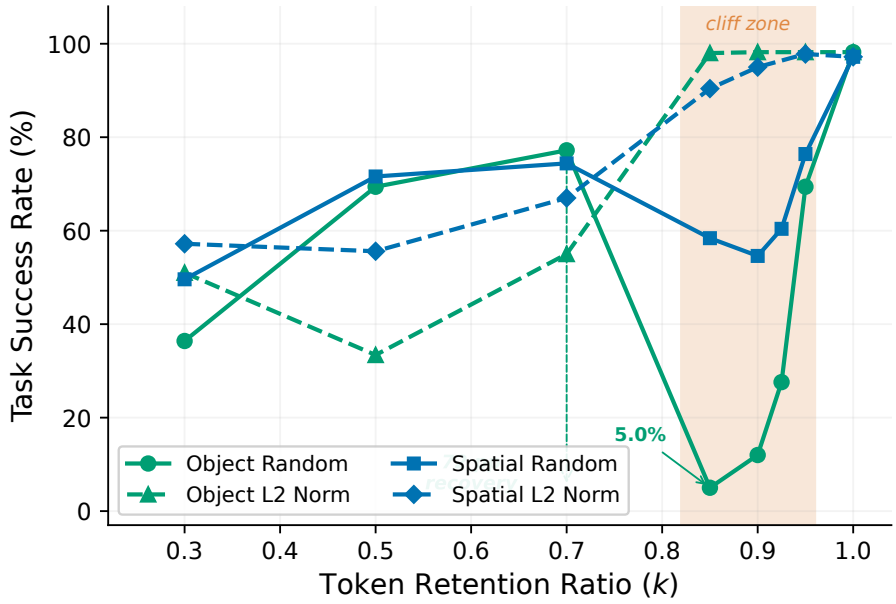


Figure 3: $\pi_0.5$ cross-suite pruning profiles. Object random (green solid) exhibits a deep cliff at $k \in [0.85, 0.925]$ with 72 pp recovery, absent on Spatial (blue solid). L2 norm (dashed) maintains a plateau on both suites through the cliff zone.

spatial discrimination. The four affected tasks all involve small, visually similar grocery items, while the robust tasks (alphabet_soup, salad_dressing) feature distinctive objects or high-contrast targets.

C.12 Smart Compress Per-Task Comparison

C.13 Smart Compress Multi-Seed Results

D Full Smart Compress Ablation

Table 15 reports all five Smart Compress configurations. The cross-attention module improves performance regardless of compression: $M \in \{32, 64, 128\}$ all achieve ~60% SR, exceeding the 44.8% baseline by 15–17 points ($p < 0.001$, two-proportion z-test). Only at $M = 16$ (87.5% compression) does performance decline modestly to 55.8%, still 11 points above baseline. The M=96 dip (mean 54.5% across three seeds, Table 14) may reflect an intermediate bottleneck that

[Placeholder: token importance heatmap]

Figure 4: $\pi_0.5$ token importance spatial distribution (16×16 grid, single observation). Left: L2 norm heatmap reveals a center-high, edge-low pattern (center rows 7–9: norm ∼265–267; edge rows 0, 14–15: norm ∼235–236). Right: at k=0.9, L2 norm selection (green) retains center tokens while discarding 50% of removed tokens from the outer ring; random selection (red marks) distributes deletions uniformly.

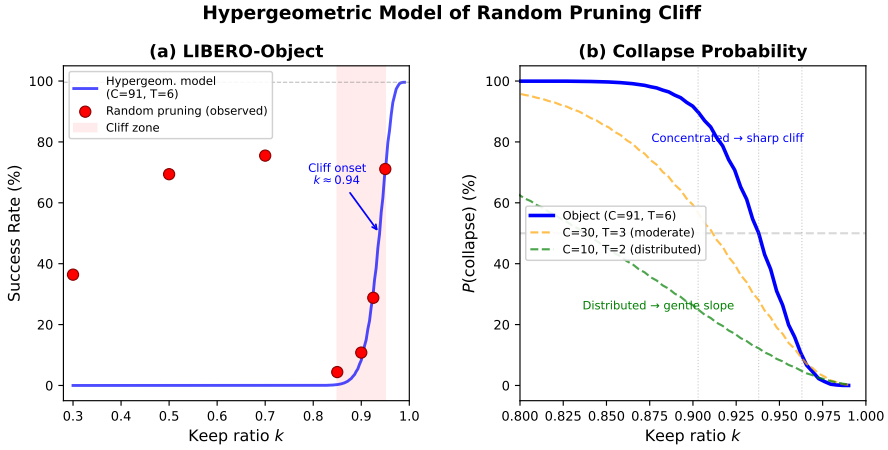


Figure 5: Hypergeometric cliff model for $\pi_0.5$ Object random pruning. (a) Model-predicted collapse probability (dashed) vs. observed success rate (solid) across the cliff zone $k \in [0.85, 0.95]$; RMSE=0.025. (b) $P(\text{collapse})$ curves for Object (C=91, concentrated) and Spatial ($C \approx 5$, distributed), illustrating why Object exhibits a sharp phase transition absent on Spatial.

introduces representational redundancy without the regularization benefit of stronger compression (M=32, M=64).

Table 16 reports performance across task suites. All suites are in-distribution: the training set (HuggingFaceVLA/libero) covers Object, Spatial, Goal, and Long. On LIBERO-Spatial, Smart Compress achieves 53.2% SR (vs. 0% under hard pruning); on LIBERO-Goal, 56.4% SR (−6.6pp from the 63.0% unpruned baseline, vs. 0% under pruning). The LIBERO-Goal decline indicates that re-encoding benefits vary across task structures even within the training distribution. Per-task results are in Table 13.

E Smart Compress Evaluation Across Task Suites

Table 16 reports the performance of Smart Compress (M=32) across three LIBERO suites. All suites are in-distribution: the training set (HuggingFaceVLA/libero) covers Object, Spatial, Goal, and Long (40 tasks, 1693 episodes total). On LIBERO-Goal, performance drops from 63.0% to

Table 10: OpenVLA success rates (%) on LIBERO-Object under random token removal across two seeds (500 episodes each). All values are within 5 pp, confirming the graded degradation curve is reproducible. Seed 7 is reported in the main text.

k	Seed 7 (%)	Seed 43 (%)	Δ (pp)
0.90	62.0	62.2	+0.2
0.70	49.0	44.4	−4.6
0.50	37.4	35.6	−1.8

Table 11: OpenVLA success rates (%) on LIBERO-Object under L2 norm-based vs. random token removal (500 episodes per configuration). Random uses seed = 7; L2 uses seed = 42. The gap widens to 4.4 pp at k=0.3 and 8.6 pp at k=0.9. Same-seed (seed42) random at k=0.3 yields 21.4% (L2 outperforms by 3.6 pp); same-seed at k=0.7 yields 44.4% (L2 outperforms by 5.4 pp). Δ is defined as L2−Random (positive indicates L2 advantage), opposite to Table 19.

k	L2 Norm SR (%)	Random SR (%)	Δ (pp)
0.95	70.6	69.4	+1.2
0.90	70.6	62.0	+8.6
0.70	49.8	49.0	+0.8
0.50	39.2	37.4	+1.8
0.30	25.0	20.6	+4.4

56.4% (−6.6pp). On LIBERO-Spatial, Smart Compress achieves 53.2% SR, where the official SmolVLA checkpoint scores 43.8% and hard pruning yields near-zero (0.6%). While 53.2% nominally exceeds the official Spatial baseline, this difference concentrates in two of ten tasks (+44pp and +48pp respectively); the remaining eight tasks show a net difference of only +0.2pp. We therefore do not treat the Spatial comparison as evidence that re-encoding improves over the uncompressed official baseline. The qualitative pattern—cliff-edge vulnerability and re-encoding recovery—extends across the three tested task structures within the training distribution. The LIBERO-Goal decline indicates that re-encoding benefits vary across task structures.

F Smart Compress Training Details

The Smart Compress module is a cross-attention pooling layer inserted between SmolVLA's pixel-shuffle vision connector and the Qwen2-0.5B language backbone. It consists of M learned query vectors that attend over the 128 vision tokens through standard multi-head cross-attention, producing M compressed tokens as input to the LLM. The compressor adds 22.2M trainable parameters (approximately 5% of SmolVLA's 451.5M total). Training is end-to-end: compressor and VLA policy parameters are jointly optimized.

The loss profile differs between configurations: $M = 32$ has higher train loss (0.11 vs. 0.045) but lower eval loss (0.052 vs. 0.062), consistent with the smaller bottleneck acting as a stronger regularizer. Despite this loss difference, both achieve identical closed-loop success rates (61.6%), reinforcing the finding that offline loss does not predict closed-loop performance (Section M).

F.1 MLP Adapter Control

To verify that re-encoding effectiveness is not specific to cross-attention, we train a parameter-matched MLP adapter as a control. The adapter is a 3-layer feed-forward network with expansion ratio 4 (hidden dimension $4 \times d_{\text{model}}$), using average pooling for token reduction from 128 to M tokens. Total parameters: 22.14M, matched to the cross-attention module's 22.2M. Training follows the identical protocol as Smart Compress: SmolVLA-Base checkpoint, 100K steps, learning rate 10^{-4}, batch size 4, same LIBERO dataset. At $M = 32$ (75% reduction), the MLP adapter achieves $63.4 \pm 2.6\%$ SR over 3 seeds, statistically indistinguishable from the cross-attention result ($60.1 \pm$

Table 12: X-VLA per-task success rates (%) at k=0.80 (40 of 50 tokens) and k=0.85 (42 of 50 tokens) on LIBERO-Object. Per-task breakdown is mean of 2 seeds (50 episodes per task per seed, 100 total); overall values are confirmed across 3 seeds ($89.0 \pm 0.7\%$ at k=0.80, $67.9 \pm 0.3\%$ at k=0.85). Despite retaining *fewer* tokens, k=0.80 outperforms k=0.85 by 21.1 pp overall. Four tasks (**bold**) suffer catastrophic interference ($\Delta \leq -50$ pp) from two additional low-importance tokens (L2 rank 41–42 of 50) at k=0.85; the remaining six are unaffected ($|\Delta| \leq 10$ pp). Cross-seed variance is ≤ 4 pp per task.

Task	k=0.80 (%)	k=0.85 (%)	Δ (pp)
alphabet_soup	100.0	100.0	0
bbq_sauce	85.0	87.0	+2
butter	**57.0**	**0.0**	**−57**
chocolate_pudding	**88.0**	**35.0**	**−53**
cream_cheese	**98.0**	**45.0**	**−53**
ketchup	**79.0**	**27.0**	**−52**
milk	85.0	95.0	+10
orange_juice	96.0	98.0	+2
salad_dressing	100.0	100.0	0
tomato_sauce	98.0	91.0	−7
Overall	**89.0**	**67.9**	**−21.1**

Table 13: Smart Compress per-task success rates on LIBERO-Object for $M \in \{32, 64, 128\}$ (seed 42; 3-seed means are reported in Table 2). All three achieve similar overall SR (59.8%–61.6%), but individual task performance varies substantially across configurations, with per-task differences up to 30 percentage points.

Task	M=32 (%)	M=64 (%)	M=128 (%)
alphabet_soup	68	74	62
bbq_sauce	46	56	34
butter	82	84	82
chocolate_pudding	90	74	82
cream_cheese	64	38	58
ketchup	66	64	62
milk	76	62	56
orange_juice	28	46	58
salad_dressing	60	64	54
tomato_sauce	36	54	50
Overall	**61.6**	**61.6**	**59.8**

1.3%; two-sample t-test, $p = 0.15$). This confirms that the re-encoding benefit does not depend on the specific module architecture.

G Offline Single-Step Action Error Analysis

Table 18 reports the single-step action prediction error under pruning, computed over 200 evaluation frames from LIBERO-Object. The unpruned model (bypass mode) produces actions with an L2 distance of 0.414 from ground truth; under ACIS pruning at k=0.50, this increases to 1.391 (3.4×). The per-dimension breakdown of the pruned-vs-unpruned difference reveals that the gripper dimension dominates (L2 = 1.02), corresponding to complete gripper state inversion on many frames. Position errors (0.20–0.30 per axis) are moderate individually but compound across the 7-dimensional action space. These single-step errors are large enough to push the robot off-trajectory within a few steps, explaining why 0% of pruned episodes succeed in closed-loop evaluation despite moderate average offline metrics.

Table 14: Smart Compress success rates across three seeds on LIBERO-Object (500 episodes each). Mean ± std reported.

M	Seed 42	Seed 43	Seed 44	Mean ± Std
32	61.6	59.2	59.6	60.1 ± 1.3
64	61.6	60.0	64.4	62.0 ± 2.3
96	52.6	58.2	52.8	54.5 ± 3.2
128	59.8	54.0	60.2	58.0 ± 3.5

Table 15: Full Smart Compress ablation on LIBERO-Object across five compression levels. M=32 and M=64 achieve similar SR (∼60%); M=96 shows a modest dip; the M=128 passthrough confirms the gain originates from re-encoding.

Method	M	Reduction	SR (%)	Episodes	Wilson 95% CI
Baseline	128	0%	44.8	500	[40.5, 49.2]
Extended training	–	–	45.0	500	[40.7, 49.3]
Hard pruning	–	50–90%	0.0	500+	[0, 0.76]
Smart Compress	128	0%	58.0	1500	±3.5
Smart Compress	96	25%	52.6	500	[48.2, 56.9]
Smart Compress	64	50%	61.6	500	[57.3, 65.8]
Smart Compress	32	75%	61.6	500	[57.3, 65.8]
Smart Compress	16	87.5%	55.8	500	[51.4, 60.1]

M=128 reports mean ± std over 3 seeds (500 episodes each). Other Smart Compress rows report single seed (seed 42) with Wilson 95% CI. See Table 14 for full multi-seed results.

H Supplementary Figures

I Attention Entropy Measurement Details

We measure the attention entropy ratio as the normalized Shannon entropy of the attention weights that the action decoder assigns to vision token positions. For a model with N vision tokens, the entropy ratio is:

$$H_{\text{ratio}} = \frac{-\sum_{i=1}^{N} \alpha_i \log \alpha_i}{\log N} \tag{1}$$

where α_i is the attention weight on the i-th vision token. The ratio ranges from 0 (all attention on one token) to 1 (uniform attention).

SmolVLA. Measured on 2400 frames from LIBERO rollouts (overall entropy) and 260 frames across all 10 LIBERO-Object tasks (per-layer analysis across 16 layers). The mean per-layer entropy ratio is 0.95 (range: 0.93 to 0.98), consistent with near-uniform attention across all 128 vision tokens. The per-layer breakdown confirms this uniformity is not an averaging artifact:

- All 16 layers: 0.93–0.98 (spread 0.05)
- Minimum: layer 13 (0.93)

The flat profile across all processing depths confirms that pixel-shuffle tokens resist selective attention at every layer: because each token encodes 16 spatially contiguous patches via channel concatenation, all tokens are comparably information-dense, and the transformer has no basis for developing spatial selectivity.

π_0. Measured on 100 frames from LIBERO-Object (tasks 20–29) across all 18 layers of the Gemma-based PaliGemma backbone (512 vision tokens: 256 per image × 2 cameras). The mean suffix-to-vision entropy ratio is 0.959 (range: 0.931–0.983, std 0.013), near-uniform across all layers with no progressive sharpening:

- All 18 layers: 0.931–0.983 (spread 0.052)

Table 16: Smart Compress (M=32) evaluation across LIBERO task suites. All suites are in-distribution (training covers Object, Spatial, Goal, Long). Hard pruning (any method, k=0.50) produces 0% or near-zero on all suites.

Suite	Baseline (unpruned)	Smart Compress M=32	Hard Pruning k=0.50
Object	44.8%	61.6%	0%
Spatial	51.8%	53.2%	0.6%
Goal	63.0%	56.4%	0%

Table 17: Smart Compress training hyperparameters. All configurations ($M \in \{16, 32, 64, 96, 128\}$) use identical procedures; only the number of query vectors differs.

Hyperparameter	Value
Model	
Base checkpoint	SmolVLA-Base (pretrained, before LIBERO fine-tuning)
Compressor architecture	Cross-attention, 2 layers, 12 heads, dropout 0.1
Compressor parameters	22.2M
Total trainable parameters	122.1M / 472.2M (25.85%)
Optimization	
Optimizer	AdamW (β_1=0.9, β_2=0.95, weight decay 10^{-10})
Learning rate	10^{-4}
Schedule	Linear warmup (1K steps, start factor 0.01) $\rightarrow$ cosine annealing ($\eta_{\min}$=2.5×10^{-6})
Max gradient norm	10.0
Total training steps	100,000
Data & batching	
Dataset	HuggingFaceVLA/libero (40 tasks, 1693 episodes across Object, Spatial, Goal, Long suites)
Batch size	4 (no gradient accumulation)†
Num workers	4
Eval interval	every 5K steps
Checkpoint interval	every 1K steps
Hardware & runtime	
Hardware	1$\times$ NVIDIA RTX 4090 D (24GB)
Precision	fp32 (no mixed precision)
Wall clock	$\sim$6h 43m per configuration
Training speed	4.1 samples/sec
Seed	42 (seeds 43, 44 for multi-seed runs)
$M = 32$ *final metrics*	
Train action loss	0.11
Eval action loss	0.052
$M = 64$ *final metrics*	
Train action loss	0.045
Eval action loss	0.062

†The official SmolVLA checkpoint uses batch size 64 across 8 GPUs (effective batch size 512). Our batch size 4 on a single GPU contributes to the lower baseline SR (44.8% vs. 87.3%). All Smart Compress configurations and the extended training control use this same batch size for fair comparison.

- Minimum: 0.931; Maximum: 0.983

The flat entropy profile, combined with π_0's extremely low CV (0.038), is consistent with its high tolerance to pruning: tokens carry near-equal importance and receive near-equal attention. Removing any subset approximates uniform spatial downsampling, preserving a proportional share of the visual field without disrupting the attention distribution.

OpenVLA. Measured on 250 frames from real LIBERO rollouts. The overall entropy ratio is 0.64. The layer-wise breakdown reveals progressive sharpening:

Table 18: Single-step action prediction error (SmolVLA, k=0.50, ACIS, zero-mask, 200 LIBERO-Object evaluation frames). Pruning increases action error by $3.4\times$ and introduces catastrophic gripper inversions.

Metric	Unpruned	Pruned (k=0.50)
Action L2 vs. ground truth	0.414	1.391
Pruned vs. unpruned L2		1.284
Relative action error	–	111.5%
Per-dimension L2 (pruned vs. unpruned)		
Position (x, y, z)	–	0.20, 0.25, 0.30
Rotation (r_x, r_y, r_z)	–	0.01, 0.02, 0.01
Gripper	–	1.02

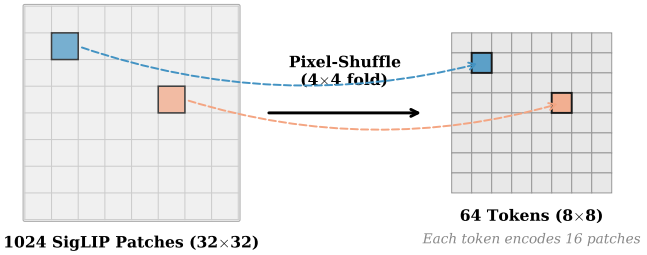


Figure 6: Pixel-shuffle folding: each 4×4 block of SigLIP patches is channel-concatenated into one token. Removing any token eliminates its receptive field (gray) with no redundancy.

- Shallow layers (0–1): 0.95–0.98
- Early layers (2–4): 0.70–0.80
- Mid layers (12–20): 0.57–0.66
- Deep layers (24–30): 0.48–0.57 (minimum 0.48 at layer 29)

12.1% of total attention weight falls on vision tokens overall, and within that fraction, the distribution is non-uniform in deeper layers. This progressive concentration indicates that OpenVLA has learned to selectively attend to task-relevant spatial locations, a pattern enabled by the MLP projector's preservation of per-patch spatial granularity.

Measurement conditions. All models are measured on real LIBERO rollout data: SmolVLA on 2400 frames (overall entropy) and 260 frames (per-layer analysis), π_0 on 100 frames from LIBERO-Object, OpenVLA on 250 frames from 50 episodes $\times$ 5 initial frames. Sample size differences reflect computational constraints (OpenVLA's 7B forward pass is substantially more expensive), but all measurements use task-relevant visual inputs. Entropy statistics show low variance across frames (std < 0.02 for OpenVLA, std $= 0.013$ for π_0).

J Connector Architecture and Token Independence

The seven-model taxonomy (Table 1) spans a continuous spectrum of token independence structures, each associated with the vision-to-LLM connector rather than model scale or action head design.

Pixel-shuffle connectors create maximal entanglement. SmolVLA's 4×4 spatial folding packs 16 patches into each token by channel concatenation (CV $= 0.314$, the highest in our study). The folding is fixed and irreversible: removing one token eliminates an entire receptive field, and no downstream layer can recover the discarded content. Tokens encode non-overlapping spatial regions with heterogeneous visual content, making every token simultaneously essential and non-redundant.

Pure linear projections produce the opposite extreme. π_0 maps each SigLIP patch through a single linear layer with no cross-token interaction, yielding CV $= 0.038$ and near-maximal entropy (0.959).

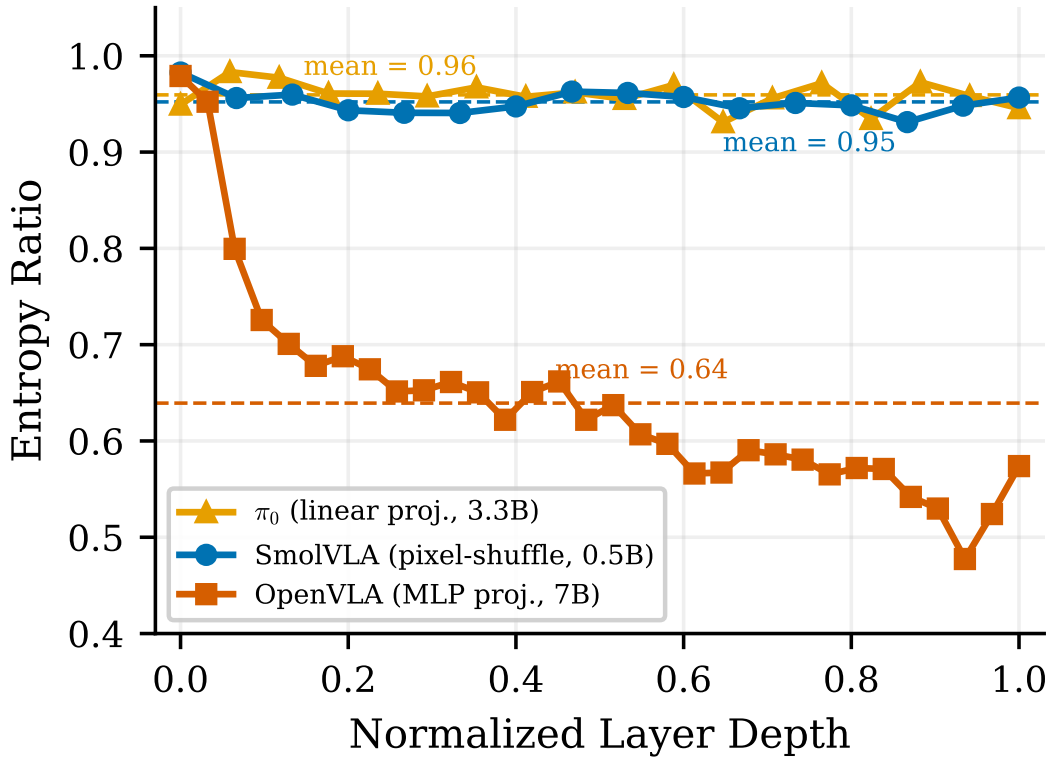


Figure 7: Per-layer attention entropy ratio for SmolVLA (16 layers), OpenVLA (32 layers), and π_0 (18 layers). SmolVLA and π_0 maintain near-uniform attention throughout, while OpenVLA progressively sharpens from 0.98 to 0.48. Despite similar entropy, SmolVLA and π_0 exhibit opposite pruning outcomes due to token importance heterogeneity (CV $= 0.314$ vs. 0.038). X-VLA (entropy $= 0.614$, CV $= 0.151$) occupies an intermediate position, consistent with its threshold-graded pattern (onset at k=0.85).

Tokens are genuinely interchangeable: any subset preserves the same per-patch information density as any other subset of equal size.

$\pi_0.5$ reveals a subtlety that π_0's offline-only evaluation conceals. The $\times\sqrt{d}$ learned scaling preserves near-uniform norms (CV $= 0.055$), yet the closed-loop response is strongly method-dependent: random removal at k=0.5 retains 69.4% SR while L2 norm-based removal collapses to 33.4%. The 36 pp gap arises because L2 ranking in a low-CV distribution selects spatially clustered tokens whose slight norm variations correlate with image regions, destroying the spatial coverage that this architecture requires. Token position analysis confirms this mechanism: at k=0.5, L2 selection produces a single dominant cluster containing 100% of retained tokens with only 78% spatial coverage, whereas k=0.7 yields two clusters with 97% in the largest and 100% coverage (Appendix O).

X-VLA's 12-layer BART encoder [44] introduces cross-token mixing through self-attention, creating partial dependencies that place it between the linear and MLP extremes (CV $= 0.151$, entropy $= 0.614$, the lowest among models). The resulting behavior is threshold-graded: performance holds through k=0.9 before a sharp cliff at k=0.85 (-31 pp). The cliff onset is reproducible across three seeds (std $= 0.3$ pp), confirming it as an architectural property.

MLP projectors (OpenVLA, OFT) apply per-patch non-linear transformations that create partial importance differentiation without cross-token entanglement. OpenVLA's CV (0.215) and OFT's CV (0.219) match within measurement noise, as do their entropy values (0.661 vs. 0.667); the connector, not the action head, is associated with token importance structure. CogACT [24] corroborates the graded pattern from the literature (74.2% SR under 78% pruning).

Parameter count does not predict pruning tolerance. Ordered by scale: SmolVLA (450M) $<$ X-VLA (880M) $<$ π_0 (3.3B) $<$ $\pi_0.5$ (4B) $<$ OpenVLA (7B). Ordered by tolerance, SmolVLA remains the minimum, but π_0 and $\pi_0.5$ at 3.3–4B surpass OpenVLA at 7B, and X-VLA at 880M tolerates more aggressive pruning than OpenVLA. The CV spectrum (0.038 to 0.314) tracks connector complexity rather than parameter count across a 16$\times$ scale range.

K CV as Diagnostic Metric: Extended Analysis

Token importance CV correlates with both the tolerance class of a model and the optimal pruning strategy, a correlation that emerges from a single mechanistic principle: CV measures the strength of the importance gradient that scoring methods can exploit.

Low-CV models have near-uniform L2 norms. L2 norm ranking in this regime captures spatial artifacts (slight norm variations that correlate with image position) rather than genuine importance differences. Selection based on these artifacts concentrates retained tokens on spatially correlated patches, destroying the uniform coverage that low-CV architectures depend on. Random selection preserves spatial uniformity by construction. The mechanism predicts that the random-over-L2 advantage should grow as CV decreases, and the data in Table 19 confirm this at two retention ratios. At k=0.5: $\Delta = +36.0$ pp ($\pi_0.5$, CV $= 0.055$) $\rightarrow +26.9$ pp (X-VLA, CV $= 0.151$) $\rightarrow -1.8$ pp (OpenVLA, CV $= 0.215$). At k=0.7: the same monotonic ordering holds ($+22.2$ pp $\rightarrow +6.5$ pp $\rightarrow -5.4$ pp). The pattern persists across a 2$\times$ change in absolute token budget, strengthening the interpretation from a coincidence at a single operating point to a structural property of the connector.

High-CV models present a genuine importance gradient where L2 norm ranking aligns with actual information content. OpenVLA (CV $= 0.215$) sits near the crossover: L2 outperforms random by 8.6 pp at k=0.9 where the gradient is most exploitable, but the advantage shrinks with decreasing k (5.4 pp at k=0.7, 1.8 pp at k=0.5) as the proportion of removed tokens exceeds the gradient's discriminative range.

Extreme compression introduces a partial reversal. At k=0.3, $\pi_0.5$'s L2 selection achieves $53.2 \pm 3.1\%$ SR (2 seeds) vs. random's 36.4%. When very few tokens remain, even spatially clustered high-norm tokens provide more signal than a sparse random sample. The V-shaped L2 curve is confirmed across seeds (σ=0.1 pp at the k=0.5 valley). The reversal does not invalidate the CV diagnostic; it adds a budget-dependent interaction that future work should characterize across more models.

Attention entropy provides a mechanistic supplement that resolves an apparent paradox. SmolVLA's entropy (0.952) is nearly as high as π_0's (0.959), yet SmolVLA collapses under any pruning. SmolVLA's tokens are uniformly *attended* but carry heterogeneous *content* due to pixel-shuffle folding. Entropy captures how the LLM allocates attention across tokens; CV captures the input-side importance structure. Entropy alone cannot distinguish "interchangeable" (π_0) from "entangled" (SmolVLA), but CV separates them cleanly (0.038 vs. 0.314). OpenVLA's progressive entropy sharpening from 0.98 to 0.48 across layers (Figure 7) is consistent with a graded regime where the LLM progressively concentrates on higher-importance tokens.

L Why Re-Encoding Succeeds: Extended Analysis

Learned re-encoding overcomes pixel-shuffle failure because it breaks the rigid spatial entanglement that the 4×4 folding creates. Cross-attention pooling integrates information across receptive field boundaries, redistributing content from the original 128 tokens into M compressed representations no longer tied to fixed spatial blocks.

The M=128 passthrough configuration suggests that re-encoding, not compression, is the key factor. Passthrough adds parameters without reducing token count, yet gains 13 pp over the baseline (44.8% $\rightarrow$ 58.0%). The relatively stable M-curve (M=16: 56.9% closely matches M=128: 58.0%, within 1.1 pp) shows that once entanglement is broken, token count is secondary.

A parameter-matched MLP adapter (22.14M parameters) achieves 63.4% SR at M=32, statistically indistinguishable from cross-attention ($p = 0.15$). Re-encoding through any learned mapping that integrates across token boundaries suffices; the attention mechanism itself is not essential.

The remedy is specific to the pathology. VTR applied to OpenVLA yields 0% SR even in passthrough mode, confirming that re-encoding disrupts rather than assists architectures whose tokens already carry independent per-patch information. VTR succeeds precisely on architectures whose rigid entanglement demands learned redistribution.

M Offline Metrics Underestimate Closed-Loop Failure

Offline evaluation systematically underestimates pruning-induced failure because per-step averages dilute catastrophic single-step events that individually terminate episodes.

X-VLA at k=0.3 demonstrates the gap directly: offline loss is 1.0014 (near-baseline), yet closed-loop SR drops to $22.0 \pm 0.9\%$ (-77 pp). A single gripper inversion (L2 $= 1.02$) is a minor perturbation in average action error but terminates an episode irreversibly. π_0's offline loss at k=0.1 shows a 2.4% *improvement* over the unpruned baseline (offline metrics only; no closed-loop evaluation available), a result that closed-loop evaluation may contradict if similar catastrophic events occur at deployment frequency.

$\pi_0.5$ adds a distinct form of evidence: the 36 pp gap between L2 and random removal at k=0.5 is invisible to any offline metric that averages across token subsets or selection methods. Offline evaluation would report degradation for both methods; only closed-loop evaluation reveals that the *choice* of pruning strategy, not just the pruning ratio, is associated with whether the model remains functional.

N Confounder Analysis

The seven models span three connector families but differ simultaneously in LLM backbone (PaliGemma for π_0/$\pi_0.5$, Llama-2 for OpenVLA, Qwen2 for SmolVLA), preventing clean isolation of the connector effect from backbone differences. The π_0–OpenVLA comparison (same 256 SigLIP-based tokens, different connector, backbone, and encoder fusion strategy) provides the closest available approximation, but a matched-backbone experiment remains the ideal control. This comparison involves differences in connector, backbone, and encoder fusion, limiting causal interpretation. Token count also varies across models (50 for X-VLA, 128 for SmolVLA, 256 for π_0/OpenVLA, 768 for $\pi_0.5$); the π_0–OpenVLA pair controls for this confounder within the CV analysis while still exhibiting the predicted CV-gap relationship.

The spatial clustering hypothesis for L2 failure on low-CV models is confirmed by token position analysis (Appendix O).

OpenVLA L2 and random results use different seeds (seed42 and seed7 respectively); same-seed (seed42) random replication confirms the cross-seed pattern at k=0.3 (21.4% random vs. 25.0% L2, gap 3.6 pp) and k=0.7 (44.4% random vs. 49.8% L2, gap 5.4 pp). Same-seed (seed42) random at k=0.5 yields 35.8% (L2: 39.2%, gap 3.4 pp), consistent with the cross-seed comparison in Table 19.

O Token Position Analysis

To validate the spatial clustering hypothesis for L2 norm failure on low-CV models, we analyze the spatial distribution of tokens selected by L2 norm ranking on $\pi_0.5$ across three retention ratios.

At k=0.7 (538 of 768 tokens retained): DBSCAN clustering identifies 2 spatial clusters, with 97% of retained tokens in the largest cluster and 100% spatial coverage of the original image grid. The broad coverage explains why L2 selection still achieves 55.0% SR at this retention ratio.

At k=0.5 (384 tokens retained): a single dominant cluster contains 100% of retained tokens with only 78% spatial coverage. The complete concentration into one spatial region confirms the clustering hypothesis: L2 norm ranking selects tokens from a contiguous high-norm region while abandoning 22% of the spatial field. This is the retention ratio where L2 performance reaches its minimum (33.4% SR, the V-shape valley).

At k=0.3 (230 tokens retained): 3 clusters emerge, with 86% in the largest and 78% spatial coverage. The fragmentation into multiple clusters at extreme compression is consistent with the partial L2 recovery observed at this ratio ($53.2 \pm 3.1\%$ SR): when the budget is very small, L2 selection is forced to sample from multiple spatial regions, partially restoring coverage.

These results directly confirm that L2 norm ranking on low-CV models introduces spatial bias that degrades performance, and that the severity peaks at intermediate retention ratios where concentration is maximal.

P CV Scaling Law: Full Data

Table 19: L2 norm vs. random removal at two retention ratios. The gap Δ(R$-$L2) scales monotonically with CV at both k=0.5 and k=0.7: low-CV models require random selection (uniform importance $\rightarrow$ L2 degenerates to spatial clustering), while high-CV models tolerate either strategy. The monotonic ordering holds across retention ratios despite a 2$\times$ change in token budget.

Model	CV	Tokens	L2 SR (%)	Random SR (%)	Δ(R$-$L2)
k=0.5 *(50% retention)*					
$\pi_0.5$	0.055	768	33.4	**69.4**	+36.0
X-VLA	0.151	50	43.4±0.8	**70.3**±0.9	+26.9
OpenVLA	0.215	256	**39.2**	37.4	−1.8
k=0.7 *(70% retention)*					
$\pi_0.5$	0.055	768	55.0	**77.2**	+22.2
X-VLA	0.151	50	58.6±0.4	**65.1**±0.1	+6.5
OpenVLA	0.215	256	**49.8**	44.4	−5.4

Q CV-Strategy Interaction at Extreme Compression

The monotonic CV-gap relationship (Table 19) holds at k=0.5 and k=0.7, but partially reverses at k=0.3 for $\pi_0.5$. At this extreme budget, L2 norm selection achieves $53.2 \pm 3.1\%$ SR (2 seeds) vs. random's 36.4%, favoring L2 by $\sim$17 pp. The reversal is consistent across both seeds (seed42: 51.0%, seed43: 55.4%; Table 8).

The mechanism is intuitive: when very few tokens remain (k=0.3 retains 230 of 768 tokens), even spatially clustered high-norm tokens provide more visual signal than a sparse random sample. At intermediate budgets (k=0.5, 384 tokens), random selection preserves enough spatial coverage to outperform L2's clustered selection; at extreme budgets, coverage from random selection becomes too sparse to be useful, and the concentrated signal from L2 selection dominates.

The interaction implies that the optimal pruning strategy depends on both CV and the absolute token budget. For low-CV models, the crossover point between random-preferred and L2-preferred regimes lies between k=0.3 and k=0.5. Characterizing this crossover across additional models and budgets is left to future work.

R Inference Cost-Benefit Analysis

The primary deployment benefit of Smart Compress at this model scale is memory, not latency. Reducing from 128 to 32 vision tokens shrinks the KV-cache footprint for the vision portion by

75%, a meaningful reduction for memory-constrained robotic hardware where KV-cache size can limit the number of concurrent policy instances or maximum context length.

End-to-end inference latency improves only modestly: 154.78 ms at M=32 versus 164.33 ms at full token count, a $1.06\times$ speedup (6% reduction). At M=64: 156.16 ms ($1.05\times$). The gap between token reduction and latency gain reflects SmolVLA's compact size (451.5M parameters): self-attention over 128 vision tokens is not the inference bottleneck, and the compressor's 22.2M-parameter cross-attention computation partially offsets downstream savings. Token pruning becomes a more compelling latency optimization for VLAs with larger token budgets (>256 tokens) or larger backbones (7B+ parameters), where attention cost scales quadratically with sequence length. Whether the training cost (100K steps, 6.5 hours per configuration) is justified depends on deployment context: for VLAs deployed across many tasks, the one-time training cost is amortized; for rapid prototyping, retraining is a barrier.

Comparison with the official checkpoint. Smart Compress at $M = 32$ ($60.1 \pm 1.3\%$ over 3 seeds) recovers from 0% SR under hard pruning to performance comparable to the official SmolVLA checkpoint (59.2%, CI [54.8%, 63.4%]). However, the difference is not statistically significant (two-sample t-test, $p \approx 0.72$), and the comparison should be interpreted cautiously given different training conditions (batch size 4 vs. 64, single vs. 8 GPUs). The primary contribution of VTR is recovering functional performance where all deletion methods yield 0%, not matching the official baseline.